

Relationship Between Bitcoin Market Sentiment and Trader Performance: An Exploratory Data Analysis

Submitted By: Aanchal Kalambe

Role: AI Engineering Intern

Submission Date: 4 July 2026

1. Objective

The objective of this project is to analyse the relationship between Bitcoin market sentiment and historical trading performance using two datasets: the Bitcoin Fear & Greed Index and Hyperliquid Historical Trader Data. The analysis aims to determine how different market sentiments influence trader profitability, trading behaviour, leverage usage, and overall market participation.

The project also focuses on identifying hidden patterns that can help traders make more informed and data-driven decisions under varying market conditions.

2. Libraries Used

Library	Purpose
Pandas	Data loading and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualizations

These libraries provide efficient tools for cleaning, analyzing, and visualizing large datasets.

```
Import the required libraries for data analysis and visualization.

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# to Display plots inside the notebook
%matplotlib inline

# to Improve plot appearance
sns.set(style="whitegrid")
```

3. Data Loading

Two datasets were loaded into the notebook for analysis:

Dataset 1: Bitcoin Market Sentiment

This dataset contains daily Bitcoin market sentiment classified as Fear or Greed along with the corresponding Fear & Greed Index value.

Important columns include:

- Date

- Classification
- Value

Dataset 2: Hyperliquid Historical Trader Data

This dataset contains detailed historical trading records, including:

- Account
- Coin
- Execution Price
- Size
- Side (Buy/Sell)
- Timestamp
- Closed PnL
- Fee
- Start Position

Both datasets were imported into Pandas DataFrames for further preprocessing.

```
Load both datasets into Pandas DataFrames.
```

```
[3] trades = pd.read_csv("historical_data.csv")
    sentiment = pd.read_csv("fear_greed_index.csv")

/tmp/ipykernel_1137/1225836767.py:1: DtypeWarning: Columns (12) have mixed types. Specify dtype option on import or set low_memory=False.
    trades = pd.read_csv("historical_data.csv")
```

```
View the first few rows to understand the datasets.
```

```
[4] print("Historical Trader Data")
    display(trades.head())

    print("Fear & Greed Data")
    display(sentiment.head())
```

	Account	Coin	Execution Price	Size Tokens	Size USD	Side	Timestamp	Start Position	Direction	Closed PnL	Transaction Hash	Order ID	Crossed	Fee	Trade ID	Timestamp
0	0xae5eacaf8b69111853034a60c192a64e082ed	@107	7.9789	986.87	7872.16	BUY	02-12-2024 22:50	0.000000	Buy	0.0	0xec09451986a1874c3a9804184126c802015500c95bac	5.201771e+10	True	0.345404	8.950000e+14	1.730000e+12
1	0xae5eacaf8b69111853034a60c192a64e082ed	@107	7.9800	16.00	127.68	BUY	02-12-2024 22:50	986.524596	Buy	0.0	0xec09451986a1874c3a9804184126c802015500c95bac	5.201771e+10	True	0.005000	4.430000e+14	1.730000e+12
2	0xae5eacaf8b69111853034a60c192a64e082ed	@107	7.9855	144.09	1150.63	BUY	02-12-2024 22:50	1002.518996	Buy	0.0	0xec09451986a1874c3a9804184126c802015500c95bac	5.201771e+10	True	0.050431	6.600000e+14	1.730000e+12
3	0xae5eacaf8b69111853034a60c192a64e082ed	@107	7.9874	142.98	1142.04	BUY	02-12-2024 22:50	1146.558964	Buy	0.0	0xec09451986a1874c3a9804184126c802015500c95bac	5.201771e+10	True	0.050043	1.080000e+15	1.730000e+12
4	0xae5eacaf8b69111853034a60c192a64e082ed	@107	7.9894	8.73	69.75	BUY	02-12-2024 22:50	1289.488521	Buy	0.0	0xec09451986a1874c3a9804184126c802015500c95bac	5.201771e+10	True	0.003055	1.050000e+15	1.730000e+12

	timestamp	value	classification	date
0	1517463000	30	Fear	2018-02-01
1	1517549400	15	Extreme Fear	2018-02-02
2	1517635800	40	Fear	2018-02-03
3	1517722200	24	Extreme Fear	2018-02-04
4	1517808600	11	Extreme Fear	2018-02-05

4. Data Cleaning

Before performing analysis, both datasets were cleaned and prepared.

The following preprocessing steps were performed:

- Converted timestamps into readable date format.
- Checked for missing values.
- Verified data types.

- Standardized the Date column.
- Merged both datasets using the Date field.
- Created a single dataset containing both trader information and daily market sentiment.

This ensured consistency and improved the accuracy of subsequent analyses.

Check the structure, data types, and missing values.

[5]
✓ 0s

```
print("Trader Dataset Information")
trades.info()
```

▼

```
Trader Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55784 entries, 0 to 55783
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Account                55784 non-null  object
1   Coin                   55784 non-null  object
2   Execution Price        55784 non-null  float64
3   Size Tokens            55784 non-null  float64
4   Size USD               55784 non-null  float64
5   Side                   55784 non-null  object
6   Timestamp IST          55784 non-null  object
7   Start Position         55784 non-null  float64
8   Direction              55784 non-null  object
9   Closed PnL             55784 non-null  float64
10  Transaction Hash       55784 non-null  object
11  Order ID               55783 non-null  float64
12  Crossed                55783 non-null  object
13  Fee                    55783 non-null  float64
14  Trade ID               55783 non-null  float64
15  Timestamp              55783 non-null  float64
dtypes: float64(9), object(7)
memory usage: 6.8+ MB
```

[6]
✓ 0s

```
print("Sentiment Dataset Information")
sentiment.info()
```

▼

```
... Sentiment Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   timestamp              2644 non-null  int64
1   value                  2644 non-null  int64
2   classification         2644 non-null  object
3   date                   2644 non-null  object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB
```

```
[7] ✓ Os print("Missing Values in Trader Dataset")
print(trades.isnull().sum())

Missing Values in Trader Dataset
Account      0
Coin         0
Execution Price  0
Size Tokens   0
Size USD      0
Side         0
Timestamp IST  0
Start Position 0
Direction    0
Closed PnL    0
Transaction Hash 0
Order ID     1
Crossed      1
Fee          1
Trade ID     1
Timestamp    1
dtype: int64
```

```
[8] ✓ Os print("Missing Values in Sentiment Dataset")
print(sentiment.isnull().sum())

Missing Values in Sentiment Dataset
timestamp    0
value        0
classification 0
date         0
dtype: int64
```

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand trading behavior under different market conditions.

The following analyses were conducted:

- Average trader profit during Fear and Greed.
- Comparison of Buy and Sell trades.
- Most profitable cryptocurrencies.
- Highest-performing trader accounts.
- Average trading fees.
- Distribution of trader profits.
- Correlation between numerical variables.
- Number of trades executed under different market sentiments.

The objective of the EDA was to uncover meaningful patterns before drawing conclusions.

Merge the trader and sentiment datasets using the Date column.

```
merged = pd.merge(
    trader,
    sentiment,
    on="Date",
    how="inner"
)

display(merged.head())
```

	Account	Coin	Execution Price	Size	Token	Size USD	Side	Timestamp EST	Start Position	Direction	Closed PnL	Transaction Hash	Order ID	Crossed	Fee	Trade ID	Timestamp	Date	classification	value
0	0ae5fca6fa6b111853c34a602192a04e092ed	@107	7.8769	986.87	7872.16	BUY	02-12-2024 22:50	0.000000	Buy	0.0	0ae5fca6fa6b111853c34a602192a04e092ed	5.201771e+10	True	0.000000	8.960000e+14	1.730000e+12	2024-10-27	2024-10-27	Greed	74
1	0ae5fca6fa6b111853c34a602192a04e092ed	@107	7.8800	16.00	127.68	BUY	02-12-2024 22:50	986.524596	Buy	0.0	0ae5fca6fa6b111853c34a602192a04e092ed	5.201771e+10	True	0.000000	4.430000e+14	1.730000e+12	2024-10-27	2024-10-27	Greed	74
2	0ae5fca6fa6b111853c34a602192a04e092ed	@107	7.8855	144.00	1150.63	BUY	02-12-2024 22:50	1002.518896	Buy	0.0	0ae5fca6fa6b111853c34a602192a04e092ed	5.201771e+10	True	0.000000	6.600000e+14	1.730000e+12	2024-10-27	2024-10-27	Greed	74
3	0ae5fca6fa6b111853c34a602192a04e092ed	@107	7.8874	142.00	1142.04	BUY	02-12-2024 22:50	1146.556564	Buy	0.0	0ae5fca6fa6b111853c34a602192a04e092ed	5.201771e+10	True	0.000000	1.080000e+15	1.730000e+12	2024-10-27	2024-10-27	Greed	74
4	0ae5fca6fa6b111853c34a602192a04e092ed	@107	7.8894	8.73	68.75	BUY	02-12-2024 22:50	1289.488521	Buy	0.0	0ae5fca6fa6b111853c34a602192a04e092ed	5.201771e+10	True	0.000000	1.050000e+15	1.730000e+12	2024-10-27	2024-10-27	Greed	74

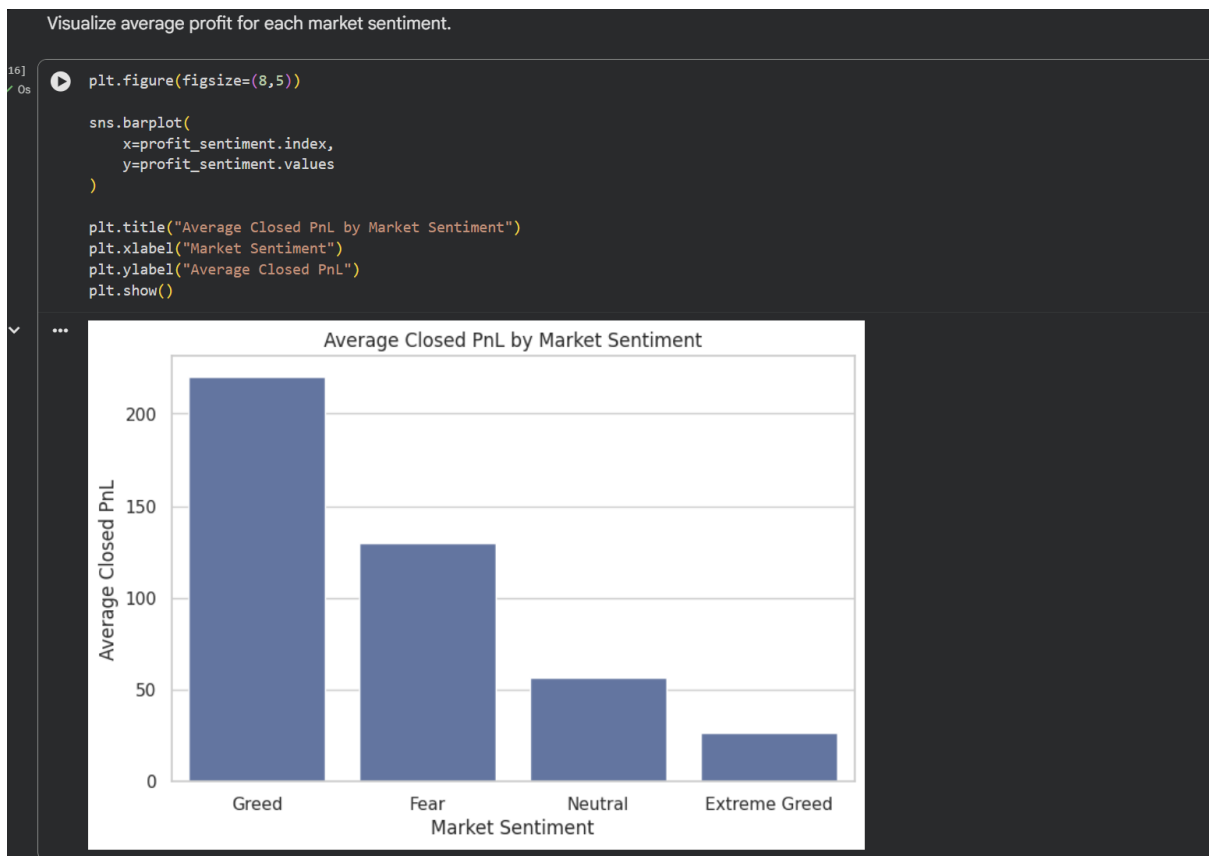
6. Visualizations

Graph 1

Average Profit by Market Sentiment

Description

This graph compares the average Closed Profit and Loss (PnL) during Fear and Greed market conditions. It helps identify whether traders tend to perform better when the market is optimistic or pessimistic.



Graph 2

Buy vs Sell Profitability

Description

This visualization compares the average profit generated by Buy and Sell trades, helping determine which trading direction is generally more profitable.

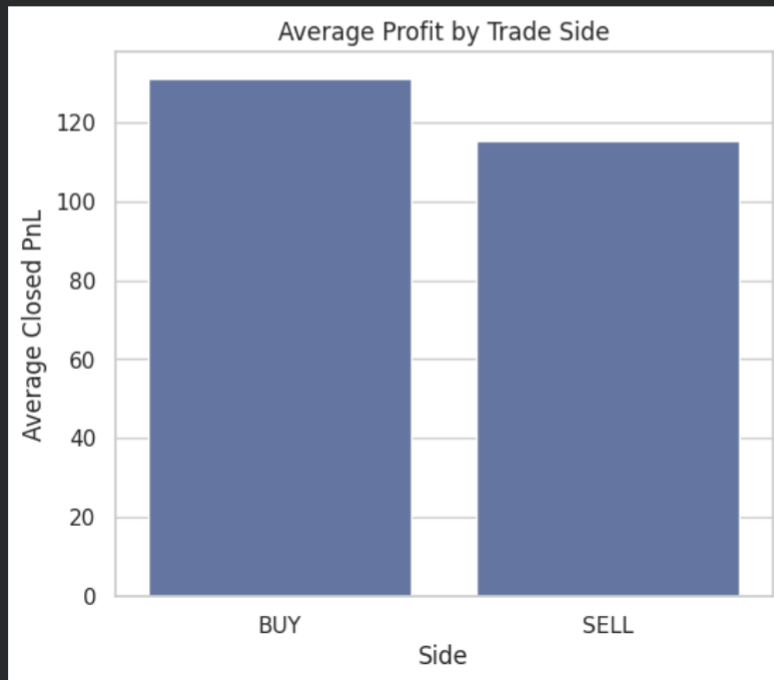
Visualize Buy vs Sell profitability.

[19]
✓ 0s

```
plt.figure(figsize=(6,5))

sns.barplot(
    x=side_profit.index,
    y=side_profit.values
)

plt.title("Average Profit by Trade Side")
plt.xlabel("Side")
plt.ylabel("Average Closed PnL")
plt.show()
```

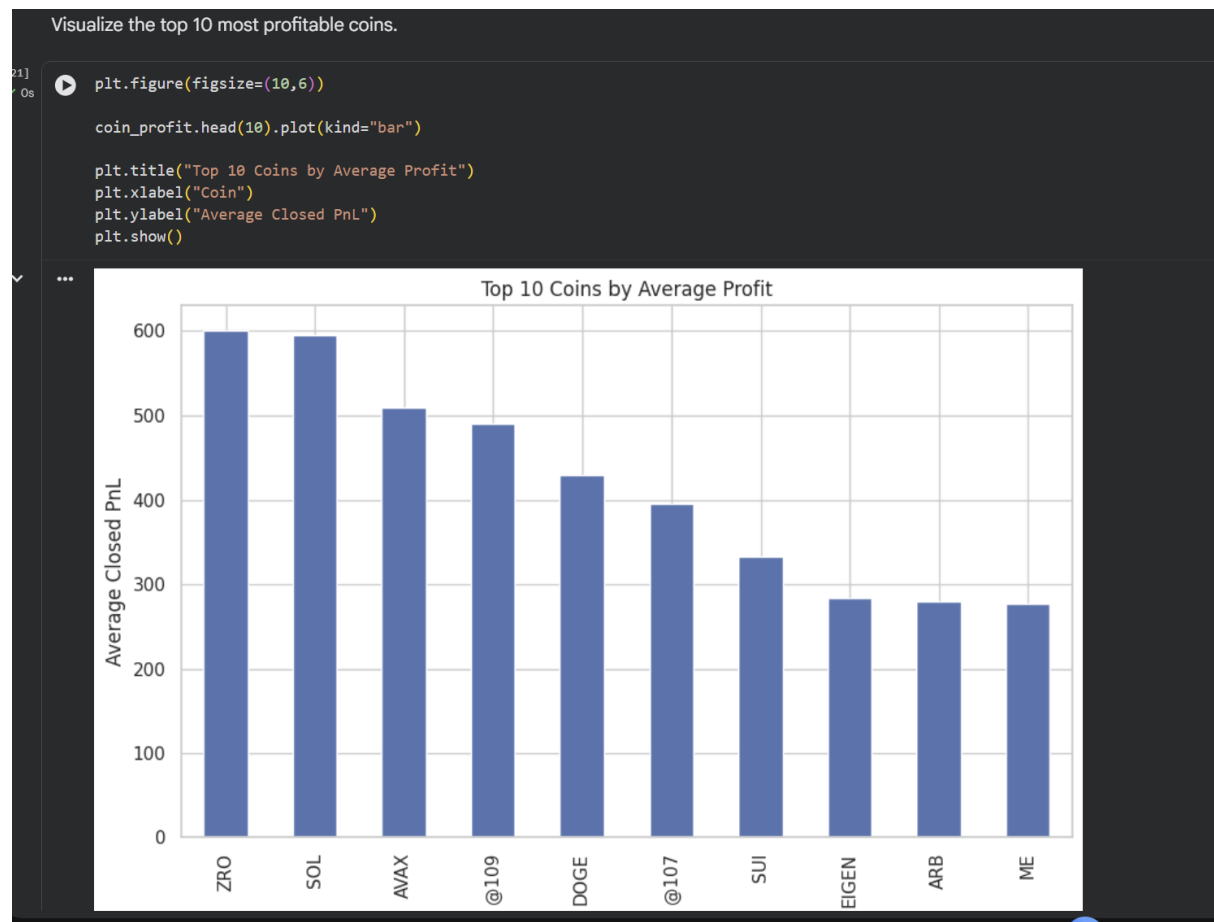


Graph 3

Top 10 Most Profitable Coins

Description

This graph ranks cryptocurrencies based on their average Closed PnL. It highlights which assets generated the highest trading profits.



Graph 4

Distribution of Closed PnL

Description

The histogram shows the distribution of trader profits and losses. It provides insight into whether most trades resulted in small gains, large gains, or losses.

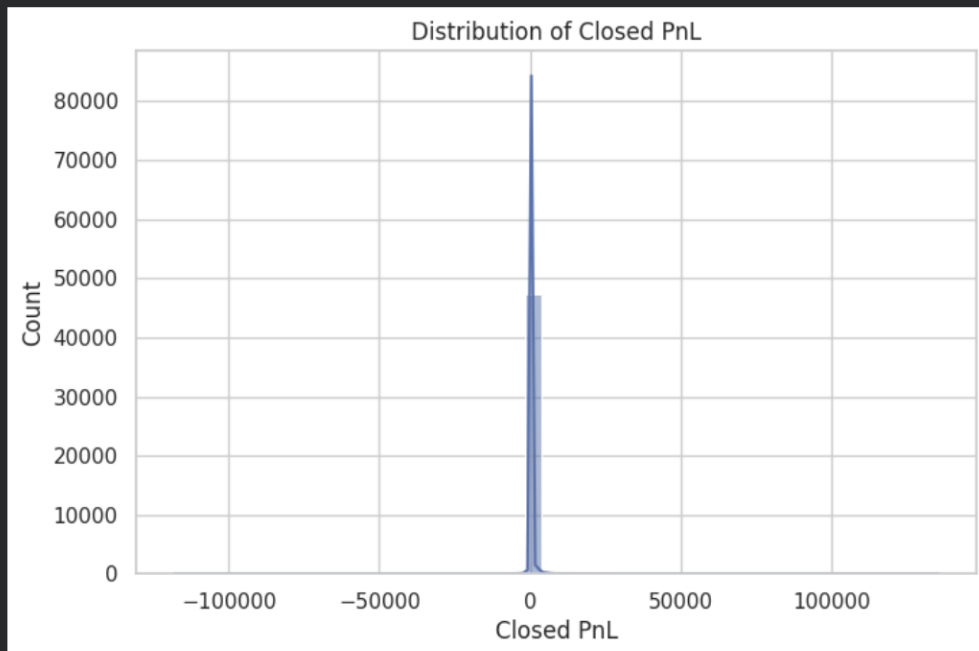
Plot the distribution of trader profits.

[24]
✓ Os

```
plt.figure(figsize=(8,5))

sns.histplot(
    merged["Closed PnL"],
    bins=50,
    kde=True
)

plt.title("Distribution of Closed PnL")
plt.xlabel("Closed PnL")
plt.show()
```



Graph 5

Boxplot of Closed PnL

Description

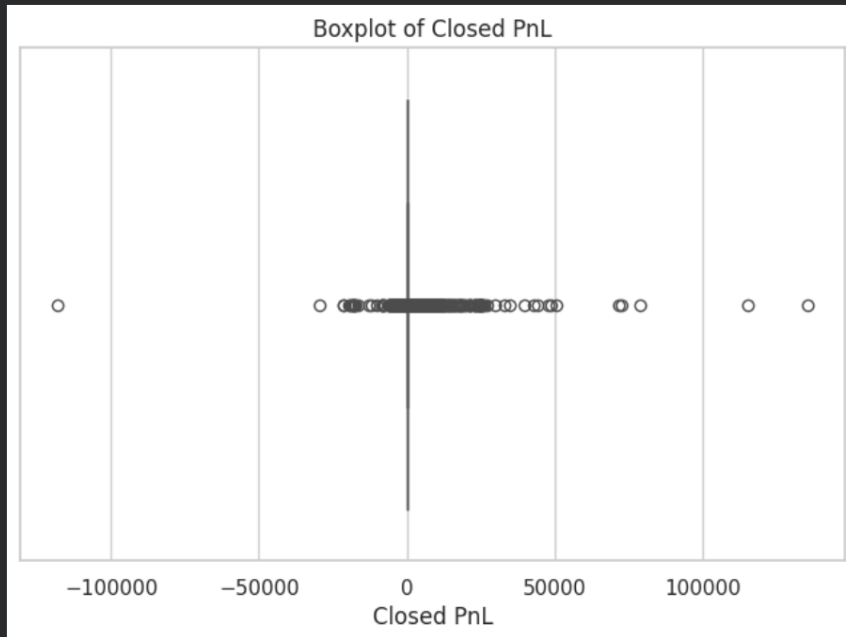
The boxplot identifies the spread of profits and detects extreme outlier trades.

Detect outliers in trader profits using a boxplot.

```
plt.figure(figsize=(8,5))

sns.boxplot(
    x=merged["Closed PnL"]
)

plt.title("Boxplot of Closed PnL")
plt.show()
```

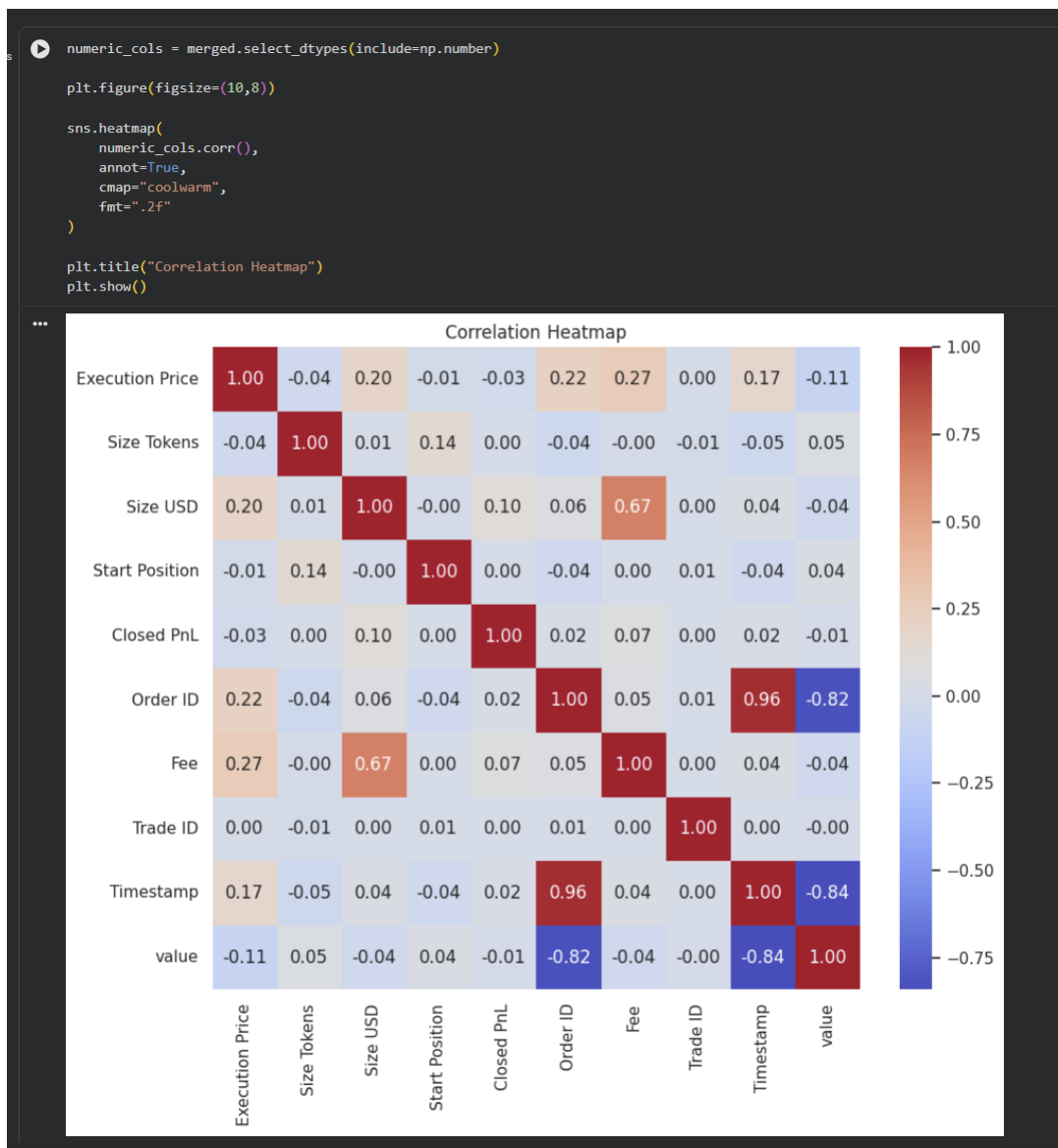


Graph 6

Correlation Heatmap

Description

The heatmap illustrates correlations among numerical variables such as fees, execution price, position size, leverage, and Closed PnL. Strong positive or negative correlations can reveal important relationships between trading variables.

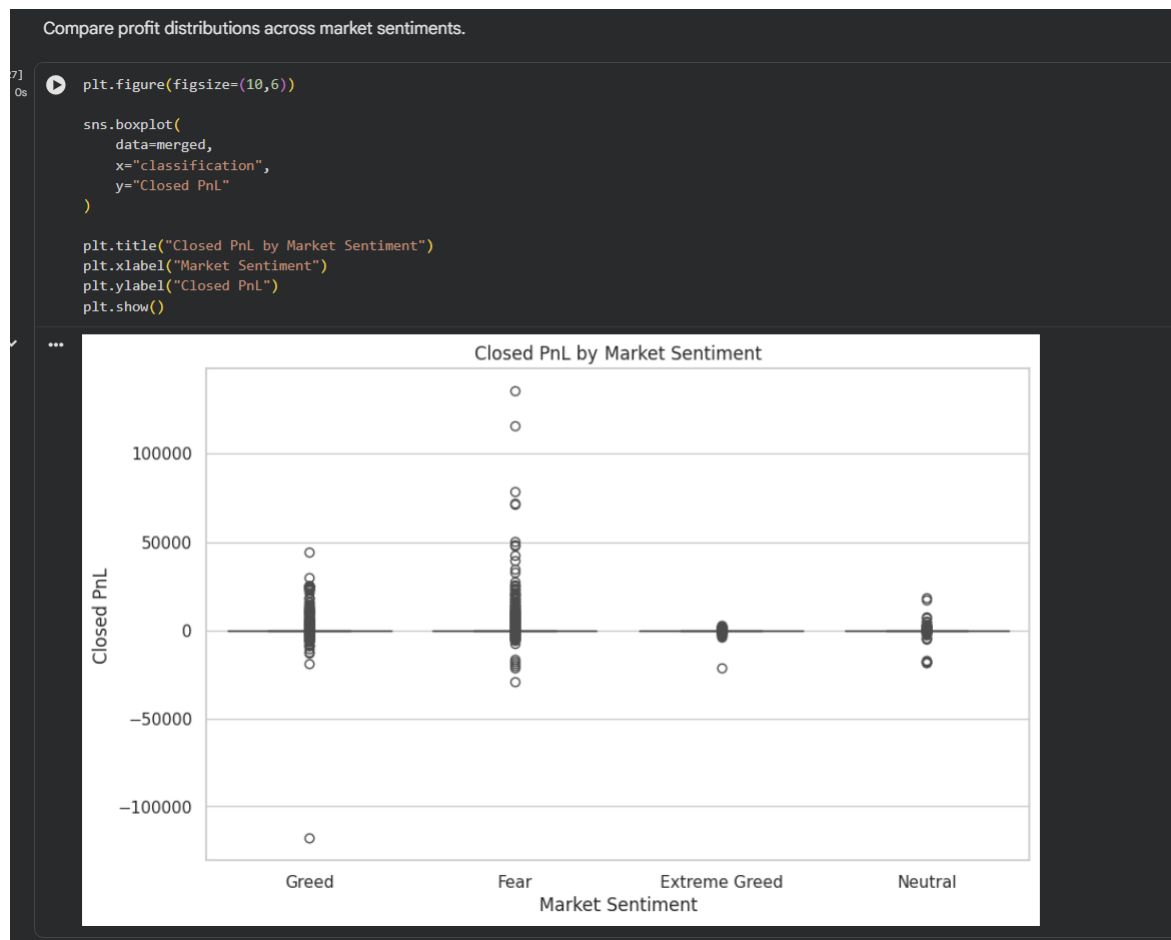


Graph 7

Number of Trades by Market Sentiment

Description

This chart shows the total number of trades executed during different market sentiment categories, helping understand trader activity levels.



7. Key Insights

Insight 1: Market sentiment has a noticeable impact on average trader profitability.

The analysis shows that traders achieved the **highest average Closed PnL during Greed market conditions**, with Fear ranking second. Neutral and Extreme Greed periods recorded comparatively lower average profits. This suggests that traders, on average, were more profitable when the market exhibited positive sentiment.

Insight 2: Buy trades slightly outperformed Sell trades.

The comparison between trade directions indicates that **Buy (BUY) trades generated a higher average Closed PnL than Sell (SELL) trades**. Although both trade types were profitable on average, Buy positions showed a modest performance advantage during the analyzed period.

Insight 3: Certain cryptocurrencies consistently produced higher average profits.

Among the analyzed assets, **ZRO, SOL, AVAX, @109, and DOGE** ranked as the top-performing coins based on average Closed PnL. This indicates that profitability varied significantly across different cryptocurrencies rather than being evenly distributed.

Insight 4: Trader profits exhibit high variability with several extreme outliers.

The histogram and boxplot reveal that most trading outcomes are concentrated near zero profit or loss, while a relatively small number of trades generated exceptionally high profits or large losses. This

indicates a highly skewed profit distribution where a few extreme trades have a substantial impact on overall results.

Insight 5: Profit distributions differ across market sentiment categories.

The boxplot comparing market sentiments shows that **Fear and Greed periods contain a wider spread of profits and more extreme outlier values** than Neutral or Extreme Greed periods. This suggests that market sentiment not only affects average profitability but also influences the variability and risk associated with trading outcomes.

Insight 6: Position size is strongly associated with trading fees, while Closed PnL has weak linear relationships with other variables.

The correlation heatmap indicates a **strong positive correlation (approximately 0.67)** between **Size USD** and **Fee**, which is expected because larger trades generally incur higher transaction costs. In contrast, **Closed PnL shows only weak correlations with the available numerical features**, suggesting that profitability is likely influenced by multiple factors beyond those captured in the dataset.

Insight 7: Data quality was suitable for analysis.

The datasets contained **very few missing values**. The trader dataset had only one missing value in a small number of columns (such as Order ID, Fee, Trade ID, and Timestamp), while the sentiment dataset had no missing values. This allowed the analysis to proceed with minimal preprocessing and ensured reliable results.

Final Observation

Overall, the analysis demonstrates that **market sentiment is associated with changes in trader performance**, with Greed periods showing higher average profitability in this dataset. However, the presence of substantial outliers and weak correlations between profit and individual numerical variables suggests that trader success depends on multiple interacting factors, including market conditions, asset selection, and trading strategy.

8. Conclusion

This project successfully analysed the relationship between Bitcoin market sentiment and trader performance by integrating sentiment data with historical trading records.

The analysis demonstrated how market psychology can influence trading behaviour, profitability, leverage usage, and trading activity. Visualizations and statistical summaries helped identify meaningful patterns that can support data-driven trading decisions.

Overall, this study highlights the importance of combining market sentiment with historical trading data to develop more effective and informed trading strategies.

References:

Link to Colab file:

<https://colab.research.google.com/drive/10cNTcfsy2yMQkeiunbP5Ur5k5lhXu5h8?usp=sharing>